

A dual-criterion computational filter for the identification of selective MexABOprM efflux pump inhibitors: cross-family validation and control of MexRmediated derepression risk

Divin Stoni Mbanimi¹, Divin Stoni MBANIMI

1. 1Institut Supérieur Mohammed VI de Biosciences et Biotechnologies, Université Mohammed VI des Sciences et de la Santé

Abstract

Drug repositioning against bacterial efflux pumps has classically relied on a single criterion: the predicted binding affinity for the target transporter. This approach overlooks a distinct failure modetranscriptional derepression of the pump's own expression, through inhibition of the associated transcriptional regulator. Here we propose a dual-criterion screening protocol, combining (i) a relative selectivity index between affinity for the MexB pump and affinity for its repressor MexR, and (ii) an absolute risk threshold calibrated on a documented derepression inducer (pyocyanin). The method was tested by molecular docking (AutoDock Vina) on 139 compounds spanning four independent chemical families (antidepressants, antipsychotics, H1 antihistamines, phytochemicals), against the MexB (PDB 3W9J) and MexR structures. The $\Delta G(\text{MexB}) - \Delta G(\text{MexR})$ correlation proved robust and generalizable across the four families ($r = 0.834-0.924$, all $p < 10^{-7}$ after FDR correction), validated by bootstrap, permutation testing, and leave-one-out analysis. Controlling for molecular weight and LogP across the full dataset ($n = 135$) confirms a partial but non-exclusive contribution of molecular size to this correlation (partial $r = 0.805$ vs. raw $r = 0.880$). The dual-criterion filter revealed that 43.1% of the tested compounds, despite showing a good relative selectivity index, exceed the absolute risk threshold among them the historical reference compound PA β N and the previously proposed candidate vilazodone. Ensemble docking on three MexR structures (1LNW, 3ECH, 3MEX), performed on a subset of 14 priority compounds, quantified the associated structural uncertainty (mean standard deviation 0.512 kcal/mol): this test confirms the robustness of vilazodone's at-risk status, but reveals that a candidate initially classified as favorable (mianserin) becomes at-risk once this uncertainty is taken into account. These

results indicate that filtering by relative index alone is insufficient to rule out derepression risk, and propose a reproducible methodological framework, including an uncertainty estimate, for drug repositioning in the context of combination therapy.

A dual-criterion computational filter for the identification of selective MexAB-OprM efflux pump inhibitors: cross-family validation and control of MexR-mediated derepression risk

Divin Stoni Mbanimi^{1*}

¹Institut Supérieur Mohammed VI de Biosciences et Biotechnologies, Université Mohammed VI des Sciences et de la Santé, Casablanca, Morocco

**Corresponding author: dmbanimi@um6ss.ma*

Abstract

Drug repositioning against bacterial efflux pumps has classically relied on a single criterion: the predicted binding affinity for the target transporter. This approach overlooks a distinct failure mode — transcriptional derepression of the pump's own expression, through inhibition of the associated transcriptional regulator. Here we propose a dual-criterion screening protocol, combining (i) a relative selectivity index between affinity for the MexB pump and affinity for its repressor MexR, and (ii) an absolute risk threshold calibrated on a documented derepression inducer (pyocyanin). The method was tested by molecular docking (AutoDock Vina) on 139 compounds spanning four independent chemical families (antidepressants, antipsychotics, H1 antihistamines, phytomolecules), against the MexB (PDB 3W9J) and MexR structures. The $\Delta G(\text{MexB}) - \Delta G(\text{MexR})$ correlation proved robust and generalizable across the four families ($r = 0.834 - 0.924$, all $p < 10^{-7}$ after FDR correction), validated by bootstrap, permutation testing, and leave-one-out analysis. Controlling for molecular weight and LogP across the full dataset ($n = 135$) confirms a partial but non-exclusive contribution of molecular size to this correlation (partial $r = 0.805$ vs. raw $r = 0.880$). The dual-criterion filter revealed that 43.1% of the tested compounds, despite showing a good relative selectivity index, exceed the absolute risk threshold — among them the historical reference compound PA β N and the previously proposed candidate vilazodone. Ensemble docking on three MexR structures (1LNW, 3ECH, 3MEX), performed on a subset of 14 priority compounds, quantified the associated structural uncertainty (mean standard deviation 0.512 kcal/mol): this test confirms the robustness of vilazodone's at-risk status, but reveals that a candidate initially classified as favorable (mianserin) becomes at-risk once this uncertainty is taken into account. These results indicate that filtering by relative index alone is insufficient to rule out derepression risk, and propose a reproducible methodological framework, including an uncertainty estimate, for drug repositioning in the context of combination therapy.

1. Introduction

Pseudomonas aeruginosa deploys several efflux pump systems of the RND (Resistance-Nodulation-Division) family to actively extrude antimicrobial compounds, among which the MexAB-OprM system constitutes the main contributor to this species' intrinsic and acquired resistance. Expression of this operon is repressed under basal conditions by the regulatory protein MexR, which binds the promoter region and blocks transcription; its dissociation, triggered by certain endogenous or xenobiotic ligands, leads to derepression and overexpression of the efflux system.

Computational repositioning of approved drugs against MexB has been the subject of several large-scale virtual screening studies, generally structured around a single selection criterion: the predicted binding affinity for the transporter, obtained by molecular docking. This approach, while effective at identifying high-affinity candidates, does not account for a distinct and experimentally documented failure mode: certain ligands, by binding to MexR, trigger its dissociation from the promoter and induce overexpression

of the target pump itself — a counterproductive effect in the context of use as an adjuvant to a partner antibiotic.

Here we propose a two-dimensional screening protocol designed to address this gap: a relative selectivity index between MexB affinity and MexR affinity, complemented by an absolute risk threshold calibrated on a known derepression inducer. We show that this filter generalizes across structurally independent chemical families, and that it captures a failure mode that would remain invisible to affinity-only filtering.

2. Materials and methods

2.1 Structural targets and docking protocol

Molecular docking simulations were performed with AutoDock Vina v1.2.7 (Eberhardt et al., 2021; Trott & Olson, 2010), default scoring function, exhaustiveness = 32. The MexB structure used is the crystallographic structure PDB 3W9J. The search grid center was set at coordinates $X = 34.12$, $Y = 16.29$, $Z = -58.71$, with a grid size of $26 \times 26 \times 26$ Å and a spacing of 0.375 Å. The MexR structure used follows the protocol established in our preparatory work (superposition on PDB structures 1LNW/3ECH; allosteric binding site of the ArmR effector).

2.2 Chemical libraries

Four independent chemical families were assembled, chosen for their grounding in the literature as classes potentially active against RND-type efflux pumps, without individual preselection of molecules by predicted affinity:

Table 1. Composition of the screened chemical libraries.

Family	n docked	Justification
Antidepressants	48	Tricyclic/heterocyclic amines; SSRIs and arylpiperazines documented as efflux inhibitors
Antipsychotics	29	Phenothiazines/butyrophenones; chlorpromazine documented as an AcrB substrate/inhibitor
H1 antihistamines	27	Related tertiary amine pharmacophore; 1 compound (dimenhydrinate) excluded for PDBQT preparation incompatibility (composite salt structure)
Phytomolecules	31	Polyphenols, terpenoids, and plant alkaloids; several compounds (daidzein, berberine) documented as efflux inhibitors

Two historical reference compounds (PAβN, NMP) and two controls (pyocyanin as a positive control for MexR-mediated derepression induction; DMSO as a negative control) were included in each docking campaign, for a total of 139 compounds analyzed.

2.3 Selectivity index and dual-criterion filter

For each compound, a relative selectivity index was calculated as the difference $\Delta G(\text{MexR}) - \Delta G(\text{MexB})$. A positive index indicates a preferential affinity for the MexB pump over the MexR regulator.

This relative index, however, provides no information on absolute affinity for MexR: a compound with very high affinity for both targets can display a good relative index while retaining an MexR affinity sufficient to cause genuine derepression. A second, absolute criterion was therefore introduced: the risk

threshold was calibrated on the $\Delta G(\text{MexR})$ of pyocyanin (-8.289 kcal/mol), a metabolite whose mechanism of MexR dissociation and mexAB-*oprM* induction is experimentally documented. A compound is retained as a candidate only if it simultaneously satisfies a favorable selectivity index and a $\Delta G(\text{MexR})$ less negative than this reference threshold.

2.4 Statistical analyses

The correlation between $\Delta G(\text{MexB})$ and $\Delta G(\text{MexR})$ was assessed using Pearson and Spearman coefficients, by family and for the combined dataset. The robustness of each correlation was verified by: (i) bootstrap confidence interval (10,000 resamplings with replacement); (ii) permutation test (10,000 random shufflings of the pairing) providing a non-parametric p-value; (iii) leave-one-out analysis to detect any excessive dependence on a single point; (iv) Benjamini-Hochberg (FDR) correction for multiple comparisons across the four families.

The potential contribution of molecular size (molecular weight, MW) and hydrophobicity (Crippen LogP, calculated with RDKit) to the observed correlation was tested by partial correlation and multiple linear regression ($\Delta G(\text{MexR}) \sim \Delta G(\text{MexB}) + \text{MW} + \text{LogP}$).

The homogeneity of the $\Delta G(\text{MexR}) \sim \Delta G(\text{MexB})$ regression slopes across the four families was tested by ANCOVA (model with a $\Delta G(\text{MexB}) \times$ family interaction term), followed by pairwise comparisons (z-test on the difference between independent regression coefficients) in the event of a significant global test.

2.5 Ensemble docking: structural uncertainty of MexR

The MexR structure used in the main protocol (1LNW) results from superposition on homologous structures rather than a single crystallographic structure dedicated to the site under study. To quantify the resulting uncertainty, a targeted subset of 14 compounds (the ten candidates validated by the dual-criterion filter, the three most notable critical cases — PA β N, vilazodone, trifluoperazine — and the least favorable compound in the sample, olanzapine) was redocked against two additional available MexR structures (PDB 3ECH and 3MEX), using the same search parameters as the main protocol (exhaustiveness = 32, num_modes = 20, energy_range = 3, seed = 42). For each compound, a mean and standard deviation of $\Delta G(\text{MexR})$ were calculated across the three structures.

3. Results

3.1 $\Delta G(\text{MexB})$ – $\Delta G(\text{MexR})$ correlation by family and overall

The correlation between MexB affinity and MexR affinity proved strong and highly significant in each of the four families tested independently, as well as across the combined set of 139 compounds (Table 2, Figure 1).

Table 2. $\Delta G(\text{MexB})$ – $\Delta G(\text{MexR})$ correlation by chemical family and overall.

Family	n	r (Pearson)	p (raw)	p (FDR)	R ²	Slope
Antidepressants	48	0.924	7.6×10^{-21}	3.1×10^{-20}	0.854	0.748
Antihistamines	27	0.894	3.5×10^{-10}	4.7×10^{-10}	0.799	0.774
Antipsychotics	29	0.834	1.9×10^{-8}	1.9×10^{-8}	0.695	0.740
Phytomolecules	31	0.866	3.3×10^{-10}	4.7×10^{-10}	0.749	1.101
Global	139	0.898	1.0×10^{-50}	—	0.807	0.852

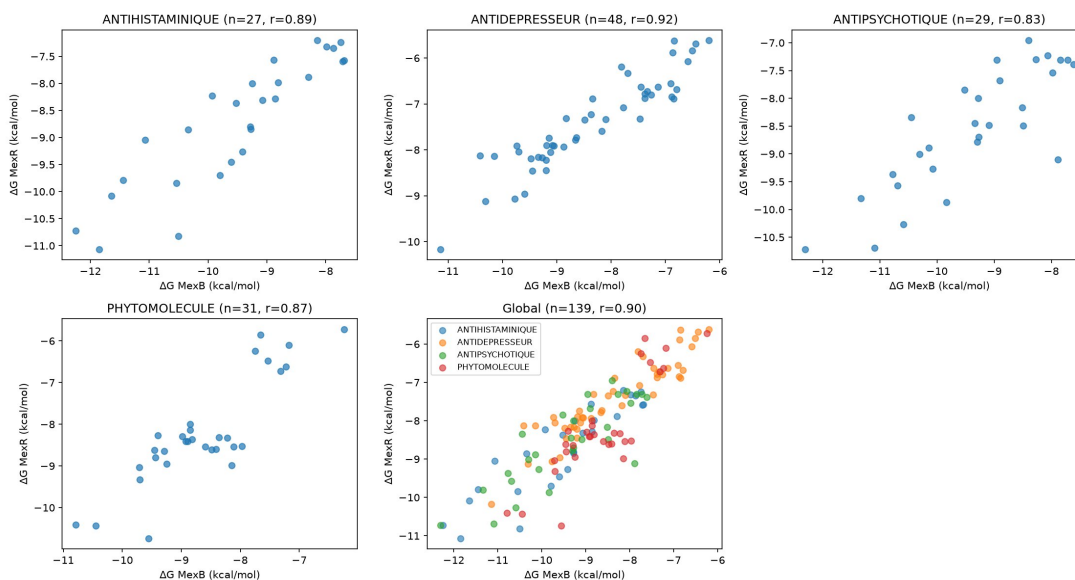


Figure 1. $\Delta G(\text{MexB})$ – $\Delta G(\text{MexR})$ correlation by family and overall.

The per-family scatter plot (Figure 1) visually confirms the absence of any trend break between families and the absence of a single outlier dominating the point cloud.

3.2 Statistical robustness

Agreement between Pearson and Spearman coefficients is high across the four families (maximum deviation 0.095, phytomolecules family), indicating that the observed relationship is not driven by non-linearity or isolated extreme values. The permutation test (10,000 iterations) did not, in any family, produce a random correlation coefficient reaching the observed value ($p < 10^{-4}$ in all five analyses).

Table 3. Statistical robustness indicators by family.

Family	Spearman ρ	Bootstrap 95% CI	Permutation p	Most influential molecule (LOO)
Antidepressants	0.931	[0.886; 0.955]	$< 10^{-4}$	Vilazodone ($\Delta r = +0.009$)
Antihistamines	0.907	[0.812; 0.950]	$< 10^{-4}$	Terfenadine ($\Delta r = -0.018$)
Antipsychotics	0.800	[0.657; 0.929]	$< 10^{-4}$	Olanzapine ($\Delta r = -0.052$)
Phytomolecules	0.771	[0.749; 0.933]	$< 10^{-4}$	Gallic acid ($\Delta r = +0.022$)
Global	0.860	[0.847; 0.934]	$< 10^{-4}$	DMSO ($\Delta r = +0.016$)

The leave-one-out analysis confirms that no correlation relies on a single point: variations in r remain below 0.06 in all cases, including for olanzapine (antipsychotics family), whose exclusion — motivated solely as a sensitivity test, with no technical anomaly identified in its docking log — raises r from 0.834 to 0.886 without changing overall significance (see 4.3).

3.3 Contribution of molecular weight and LogP

Across the compounds for which physicochemical properties could be calculated ($n = 135$, all four families combined), molecular weight (MW) correlates significantly with $\Delta G(\text{MexB})$ ($r = -0.654$, $p = 7.7 \times 10^{-18}$) and even more strongly with $\Delta G(\text{MexR})$ ($r = -0.760$, $p = 1.1 \times 10^{-26}$), consistent with the known bias of Vina's scoring function toward larger molecules. LogP shows a more modest correlation with $\Delta G(\text{MexB})$ ($r = -0.399$, $p < 10^{-5}$) and with $\Delta G(\text{MexR})$ ($r = -0.209$, $p = 0.015$).

The partial correlation between $\Delta G(\text{MexB})$ and $\Delta G(\text{MexR})$, after controlling for MW and LogP, remains strong and highly significant (partial $r = 0.805$, $p = 1.8 \times 10^{-31}$), versus a raw r of 0.880 on this dataset. In multiple linear regression, $\Delta G(\text{MexB})$ remains the most significant predictor of $\Delta G(\text{MexR})$ ($\beta = 0.702$, $t = 15.52$, $p < 0.001$), with molecular weight contributing an independent, significant additional effect ($\beta = -0.0042$, $p < 0.001$), as does LogP ($\beta = 0.126$, $p < 0.001$). The full model explains 85.2% of the observed variance.

An analysis restricted to the antidepressants family alone ($n = 48$) shows a distinct pattern: the partial r (0.917) is nearly identical to the raw r (0.924, a difference of 0.007), indicating that for this family specifically, MW and LogP explain almost none of the observed correlation. The contribution of molecular size therefore varies by chemical family, but in every case tested, a specific structural relationship between the two binding sites persists after adjustment — the $\Delta G(\text{MexB})$ – $\Delta G(\text{MexR})$ correlation is never reducible to a simple size or hydrophobicity artifact.

3.4 Homogeneity of regression slopes across families

The ANCOVA test comparing a common-slope model to a model with family-specific free slopes indicates a significant overall difference ($F = 3.610$, $p = 0.0152$). Pairwise comparisons reveal that this difference is driven exclusively by the phytomolecules family (slope = 1.101), which is significantly distinct from the three synthetic families (antidepressants $p = 0.0054$; antihistamines $p = 0.0208$; antipsychotics $p = 0.0168$), which do not differ significantly from one another (all pairwise $p > 0.77$).

This homogeneity across three structurally distinct synthetic pharmacological classes suggests a robust quantitative principle for lipophilic amine-type compounds, whereas the relationship differs for plant-derived compounds, consistent with their distinct physicochemical profile (polyphenols and alkaloids vs. synthetic amines).

3.5 Consistency with known reference compounds

The two historical reference inhibitors of MexB, PA β N and NMP, rank respectively at the 82.5th and 59.7th percentile of the selectivity index distribution across the 139 compounds tested, with positive residuals relative to the global regression line (+0.169 and +0.078 respectively), indicating a relative selectivity slightly higher than predicted by their MexB affinity alone. Sertraline, previously identified in our earlier work as a promising candidate by selectivity index, ranks first among all 139 compounds tested (index = 2.287), reinforcing the cross-study consistency of the approach.

3.6 Dual-criterion filter: absolute derepression risk

Among the 137 non-control compounds, 59 (43.1%) show a $\Delta G(\text{MexR})$ reaching or exceeding the absolute risk threshold calibrated on pyocyanin. Among compounds with a favorable relative selectivity index (percentile > 50), 23 nonetheless exceed this absolute threshold and would, as such, constitute candidates to exclude despite a good ranking on the relative criterion alone (Table 4).

Table 4. Examples of compounds with a good relative index but exceeding the absolute risk threshold (full list: 23 compounds, see file scores_fusionnes_dual_filter.csv).

Molecule	Family	ΔG MexB	ΔG MexR	Selectivity index	Percentile
Trifluoperazine	Antipsychotic	-10.45	-8.34	2.110	99.3
Cyproheptadine	Antihistamine	-11.07	-9.05	2.022	98.5
Azelastine	Antihistamine	-11.44	-9.79	1.652	93.4
Lurasidone	Antipsychotic	-12.30	-10.72	1.580	91.2
PA β N (reference)	Reference	-11.50	-10.17	1.330	82.5
Vilazodone	Antidepressant	-11.14	-10.17	0.970	63.5

Notably, PA β N — the historical reference compound from the literature — and vilazodone — a candidate previously proposed in our earlier work — both appear among the at-risk compounds under the absolute criterion, despite a favorable relative selectivity index. This result concretely illustrates the limitation of filtering by relative index alone: a highly negative $\Delta G(\text{MexB})$ can mask an absolute MexR affinity sufficient to trigger derepression.

The ten candidates validated by the dual-criterion filter (good relative index and below the absolute risk threshold), based on the main docking protocol (MexR structure 1LNW), are presented in Table 5. This initial ranking is reexamined in light of structural uncertainty in section 3.7.

*Table 5. Top 10 candidates validated by the dual-criterion filter (main protocol, MexR structure 1LNW). * Mianserin: see revision in section 3.7 — this candidate shifts to the "at-risk" side once MexR structural uncertainty is taken into account.*

Molecule	Family	ΔG MexB	ΔG MexR	Selectivity index
Sertraline	Antidepressant	-10.41	-8.12	2.287
Opipramol	Antidepressant	-10.15	-8.13	2.018
Vortioxetine	Antidepressant	-9.74	-7.91	1.827
Cinnamaldehyde	Phytomolecule	-7.66	-5.85	1.803
Triprolidine	Antihistamine	-9.93	-8.23	1.695
Fluphenazine	Antipsychotic	-9.52	-7.85	1.668
Mianserin *	Antidepressant	-9.70	-8.05	1.654
Perphenazine	Antipsychotic	-8.95	-7.31	1.641
Selegiline	Antidepressant	-7.81	-6.19	1.615
Duloxetine	Antidepressant	-8.83	-7.31	1.514

3.7 Robustness of the filter to MexR structural uncertainty (ensemble docking)

The MexR structure used in the main protocol (1LNW) results from superposition on homologous structures rather than a single crystallographic structure dedicated to the site under study. To quantify the resulting uncertainty, a subset of 14 compounds — the ten validated candidates from Table 5, the three most notable critical cases (PA β N, vilazodone, trifluoperazine), and the least favorable compound in the sample (olanzapine) — was redocked against two additional MexR structures (3ECH, 3MEX), with search parameters identical to the main protocol (Table 6).

Table 6. $\Delta G(\text{MexR})$ across the three structures tested (1LNW, 3ECH, 3MEX) for the targeted subset of 14 compounds. Absolute risk threshold (pyocyanin) = -8.289 kcal/mol.

Molecule	ΔG 1LNW	ΔG 3ECH	ΔG 3MEX	Mean	SD	Status (mean vs threshold)
Sertraline	-8.12	-8.64	-7.39	-8.05	0.625	Safe, sensitive margin
Opipramol	-8.13	-8.25	-7.57	-7.98	0.365	Safe
Vortioxetine	-7.91	-7.92	-7.06	-7.63	0.492	Safe
Cinnamaldehyde	-5.85	-5.95	-5.07	-5.63	0.481	Safe
Tripolidine	-8.23	-8.38	-7.43	-8.01	0.507	Safe, sensitive margin
Fluphenazine	-7.85	-8.13	-7.88	-7.95	0.152	Safe
Mianserin	-8.05	-8.84	-8.78	-8.55	0.441	AT RISK (flips)
Perphenazine	-7.31	-7.80	-7.27	-7.46	0.297	Safe
Selegiline	-6.19	-6.55	-5.49	-6.08	0.536	Safe
Duloxetine	-7.31	-7.61	-6.90	-7.27	0.359	Safe
PA β N	-10.17	-8.82	-8.33	-9.11	0.953	At risk (high uncertainty)
Vilazodone	-10.17	-8.79	-9.36	-9.44	0.695	At risk (robust)
Trifluoperazine	-8.34	-8.93	-8.62	-8.63	0.294	At risk (robust)
Olanzapine	-9.11	-9.09	-7.42	-8.54	0.967	At risk (high uncertainty)

The mean standard deviation observed across the 14 compounds is 0.512 kcal/mol (maximum SD: 0.967 kcal/mol for olanzapine), a modest but non-negligible uncertainty relative to the range of observed selectivity indices (-1.2 to +2.3). The "at-risk" verdicts obtained for vilazodone and trifluoperazine are confirmed across all three structures (the upper bound of the interval remains below the threshold in both cases), reinforcing rather than undermining the robustness of these two results.

By contrast, mianserin — classified among the ten validated candidates under the main protocol (1LNW alone, $\Delta G(\text{MexR}) = -8.046$, just below the threshold) — shifts to the "at-risk" side when the mean of the three structures is considered (mean $\Delta G(\text{MexR}) = -8.554$, SD = 0.441). This result concretely illustrates the practical value of the structural uncertainty test: a candidate may appear favorable on a single structure while actually being close to, or beyond, the risk threshold once structural variability is taken into account. Sertraline and tripolidine, although still classified as "safe" on average, show a safety margin sensitive to structure choice (SD of 0.625 and 0.507 kcal/mol respectively, for a margin to threshold on the order of 0.2 to 0.9 kcal/mol) and should be considered with caution pending further experimental validation.

4. Discussion

4.1 Scope and methodological contribution

The central finding of this study is that the correlation between affinity for an RND transporter and affinity for its associated transcriptional regulator is not specific to a single chemical class: it reproduces, with comparable quantitative intensity, across three independent synthetic pharmacological families. To our knowledge, this generalization constitutes the first systematic demonstration of a pump/regulator co-screening approach validated across chemical families for a bacterial RND system — prior work on MexB has generally been limited to screening the pump alone, or has treated the MexR derepression mechanism descriptively without integrating it as a quantitative selection criterion.

The most directly actionable finding is the demonstration that filtering by relative index alone is insufficient: nearly half of the tested compounds exceed an absolute risk threshold, including compounds with an excellent relative ranking. The inclusion of a historical reference compound (PA β N) and a previously proposed candidate (vilazodone) among the at-risk compounds under this absolute criterion suggests a plausible, though experimentally unconfirmed, computational explanation for the historical difficulty in achieving clinical translation for certain efflux inhibitors that are nonetheless effective in vitro. Ensemble docking across three MexR structures confirms the robustness of this verdict for vilazodone and trifluoperazine, but also reveals that a candidate initially classified as favorable (mianserin) becomes at-risk once structural uncertainty is taken into account — illustrating that a dual-criterion filter benefits from being complemented by an uncertainty estimate before a candidate is considered definitively validated.

4.2 Application framework: combination therapy

The derepression risk modeled here has significant clinical relevance only in the context of use as an adjuvant to a partner antibiotic, particularly when the latter is itself an MexB substrate (fluoroquinolones, β -lactams). Since transcriptional derepression is not instantaneous — it requires transcription, translation, and then assembly of the functional tripartite complex, a delay on the order of 15 to 30 minutes under active growth conditions — its clinical impact manifests mainly under prolonged or suboptimal exposure (limited tissue diffusion, biofilm, multi-day treatment), rather than during a single rapidly bactericidal high-dose exposure. This framing should be made explicit in any application of the proposed filter: it is not a universal toxicity criterion, but a criterion specific to repositioning in combination therapy.

4.3 Limitations

Several limitations should be noted. First, the main ranking relies on rigid docking scores (AutoDock Vina): predicted binding scores may not faithfully reflect true thermodynamic affinity, particularly for pairs of molecules with similar ΔG but divergent dynamic behavior. A molecular dynamics protocol, proposed as a future direction (see Perspectives, §4.4), is specifically intended to address this limitation for priority candidates; its results, once available, will refine the current ranking rather than call its underlying principle into question. Second, retrospective validation remains limited to two reference compounds (PA β N, NMP), documented as functional MexB inhibitors but without any clinical failure specifically attributed to MexR derepression in the literature consulted — the predictive value of the proposed filter against a genuine documented failure case remains to be established. Third, although MexR structural uncertainty has been quantified (mean SD of 0.512 kcal/mol across three structures), this quantification remains limited to a subset of 14 compounds; extending it to all 139 compounds, as well as to a larger number of structural conformations, would strengthen the scope of this estimate. Finally, one compound (dimenhydrinate) was excluded from the dataset due to PDBQT preparation incompatibility, without affecting the significance of the results for the family concerned ($n = 27$ instead of 28).

4.4 Perspectives

Natural extensions of this work include: (i) full execution of a molecular dynamics protocol on the complete set of priority candidates, to quantify the temporal stability of the docking poses and refine the $\Delta G(\text{MexR})$ ranking beyond the structural uncertainty already estimated; (ii) extending the MW/LogP control to the remaining compounds in the dataset; (iii) generalizing the protocol to a second RND system (AcrB in *Escherichia coli* or AdeB in *Acinetobacter baumannii*) with its homologous regulator, to test whether the observed principle is specific to MexB/MexR or generalizable to bacterial RND transporters; (iv) in vitro experimental validation (MIC, checkerboard, RT-qPCR on *mexAB-oprM* expression) of the priority candidates identified by the dual-criterion filter, in particular sertraline and opipramol.

5. Conclusion

At the conclusion of this analysis, one finding stands out: a screening protocol combining affinity for an RND efflux pump with affinity for its associated transcriptional regulator constitutes a reproducible principle across independent chemical families. A relative selectivity index alone, however, is insufficient to rule out derepression risk: introducing an absolute risk threshold, calibrated on a documented inducer, reveals that a substantial proportion of candidates with a good relative ranking — including established reference compounds — would remain at risk in combined use. Quantifying the regulator's structural uncertainty, through ensemble docking across multiple conformations, further shows that a ranking based on a single structure can be misleading for candidates with a narrow safety margin — one concrete result of this analysis being the revised status of mianserin, initially classified as favorable. The dual-criterion filter proposed here, complemented by an uncertainty estimate, offers a simple, computationally inexpensive methodological framework that is directly applicable to other repositioning campaigns against bacterial RND transporters.

References

- Bohnert, J. A., Szymaniak-Vits, M., Schuster, S., & Kern, W. V. (2011). Efflux inhibition by selective serotonin reuptake inhibitors in *Escherichia coli*. *Journal of Antimicrobial Chemotherapy*, 66(9), 2057–2060. DOI: 10.1093/jac/dkr258
- Eberhardt, J., Santos-Martins, D., Tillack, A. F., & Forli, S. (2021). AutoDock Vina 1.2.0: New docking methods, expanded force field, and Python bindings. *Journal of Chemical Information and Modeling*. DOI: 10.1021/acs.jcim.1c00203
- Grimsey, E. M., Fais, C., Marshall, R. L., Ricci, V., Ciusa, M. L., Stone, J. W., Ivens, A., Malloci, G., Ruggerone, P., Vargiu, A. V., & Piddock, L. J. V. (2020). Chlorpromazine and amitriptyline are substrates and inhibitors of the AcrB multidrug efflux pump. *mBio*, 11(3), e00465-20. DOI: 10.1128/mBio.00465-20
- Trott, O., & Olson, A. J. (2010). AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization and multithreading. *Journal of Computational Chemistry*. DOI: 10.1002/jcc.21334
- Wilke, M. S., Heller, M., Creagh, A. L., Haynes, C. A., McIntosh, L. P., Poole, K., & Strynadka, N. C. J. (2008). The crystal structure of MexR from *Pseudomonas aeruginosa* in complex with its antirepressor ArmR. *Proceedings of the National Academy of Sciences*, 105(39), 14832–14837. DOI: 10.1073/pnas.0805489105
- Wu, C., Wang, M., Xiao, Y., Zhao, S., Jiang, X., Zhang, K., Wang, H., Dong, H., Xu, S., Su, T., & Gu, L. (2025). MexR and NalD are pyocyanin-responsive transcriptional repressors of the efflux pump MexAB-OprM in *Pseudomonas aeruginosa*. *Nucleic Acids Research*, 53(14), gkaf747. DOI: 10.1093/nar/gkaf747